08/07/2025 (FN)

Hall Ticket Number:

MR-22

Code: B425CS4

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular/Supplementary Examinations, July 2025

Database Management Systems

(Common to CSE, IT, CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define a Database and Database Management System.	1	1	1	1
1. b)	List three levels of abstraction.	1	1	1	1
1. c)	Define a constraint and list types of constraints.	1	2	2	1
1. d)	List any two DDL commands.	1	1	2	1
1. e)	What is trigger in DBMS?	1	2	3	1
1. f)	Define Normalization.	1	2	3	1
1. g)	What is atomicity?	1	1	4	1
1. h)	List various transaction states.	1	1	4	1
1. i)	List any two types of database indices.	1	1	5	1
1. j)	What is hashing?	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the significance of conceptual design using ER Model?	5	2	1	1
2. b)	Compare and contrast file system and DBMS with respect to data access and security.	5	2	1	2
	OR				
2. c)	Illustrate mapping cardinalities and participation of entities in a relationship with suitable example.	5	2	1	1
2. d)	Explain structure of DBMS in detail.	5	2	1	1
3. a)	Illustrate table creation and enforcing integrity constraints with a suitable example.	5	2	2	1
3. b)	Explain Relational Algebra.	5	2	2	1
	OR				
3. c)	What is referential integrity? Explain the purpose of referential integrity in detail with appropriate example?	5	2	2	2
3. d)	Differentiate Tuple relational calculus and domain relational calculus.	5	2	2	1
4. a)	Consider a relation R with following attributes {SSN,ENAME, PNAME,PLOC,HOURS} where SSN is Social Security number and it is primary key, Ename is employee name, PNUMBER is project number, PNAME is project name, PLOC is project location and HOURS is number of working hours per day. Below mentioned are the decomposed relations. Apply loss less join decomposition algorithm and check whether the decomposition is lossy or loss less. R1= {SSN, ENAME} R2= {PNUMBER, PNAME, PLOC} R3= {SSN, PNUMBER, HOURS}.	5	3	3	2
4. b)	What is attribute closure? Explain how to find attribute closure with an example?	5	3	3	1

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	OR				
4. c)	List and explain aggregate operations in SQL.	5	2	3	1
4. d)	Explain various types of Join operations in SQL.	5	2	3	1
5. a)	Illustrate implementation of Atomicity and Durability properties.	5	3	4	1
5. b)	Let's say two transactions T1 and T2 want to update the same bank account balance at a time. • T1: Reads balance, adds ₹100, writes it back. • T2: Reads balance, deducts ₹50, writes it back. Analyse and critique what would happen with and without mutual exclusion.	5	4	4	2
	OR				
5. c)	Mutual exclusion is a mechanism to achieve serializability. Justify?	5	5	4	2
5. d)	Explain lock based protocols?	5	3	4	1
6. a)	Discuss file organization and importance of indexing.	5	2	5	1
6. b)	Illustrate B+ trees.	5	3	5	1
	OR				
6. c)	Demonstrate Index Sequential Access Method (ISAM).	5	3	5	1
6. d)	Explain cluster index in detail.	5	2	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome